

ASSIGNMENT #02

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Abstract

This report analyzes a regenerative Rankine cycle with an open feedwater heater (OFWH) and two turbines. Thermodynamic state properties were computed using steam-property functions (HSteam/SSteam) and the system was solved using the instructor method: each turbine/pump is first assigned an *ideal* (reversible, adiabatic) outlet by enforcing the entropy balance ($\Delta s = 0$), then the *real* outlet is computed by enforcing the appropriate isentropic efficiency equation while leaving the core balances “untouched.” The steam flow was scaled until the net power output equaled 500 kW. Key results (from the final spreadsheet) are: steam flow = 2.105 MMSCFD, bleed fraction $b = 0.12949$, thermal efficiency $\eta_{th} = 33.73\%$, maximum theoretical (Carnot) efficiency $\eta_{\max} = 76.0\%$ (using $T_H = 1250$ K and $T_0 = 300$ K), thermodynamic efficiency $\eta_{td} = 44.38\%$, and total lost work $\dot{W}_{lost,tot} \approx 626.59$ kW. Utility costing at this operating point gives a total annual cost of approximately $\$2.253 \times 10^5/\text{yr}$ (natural gas + cooling water + pump electricity).

Goals

- Determine the steam flow rate required for 500 kW net output (MMSCFD).
- Determine bleed fraction b such that the OFWH outlet is 1 °C subcooled at OFWH pressure.
- Compute thermal efficiency, maximum theoretical efficiency (as assigned), and thermodynamic efficiency.
- Compute lost work (exergy destruction) for each unit and in total.
- Compute annual costs for natural gas, cooling water, and pump electricity.
- Compute methane/air requirements and estimate CO₂ and NO_x emissions.
- Perform (or document) optimization of HP turbine discharge pressure; include method and placeholder value.

1 Background Knowledge and Governing Theory

1.1 Rankine Cycle Overview

A Rankine cycle produces net shaft work by adding heat to a pressurized liquid in a boiler, expanding the resulting steam through turbines, rejecting heat in a condenser, and pumping the condensate back to the boiler pressure. Under steady-state, steady-flow assumptions and with negligible changes in kinetic and potential energy, the specific enthalpy changes dominate the energy accounting.

1.2 Regeneration and the Open Feedwater Heater (OFWH)

Regeneration increases cycle performance by extracting a fraction of steam from an intermediate turbine pressure (“bleed” steam) and mixing it with the feedwater. In an **open** feedwater heater, the two streams mix directly at a common pressure; therefore, the OFWH is modeled as a steady-flow **mixer**. The bleed fraction is chosen so that the mixed outlet reaches a specified condition. In this homework, the OFWH outlet is 1 °C subcooled at the OFWH pressure.

1.3 Instructor “Untouched Balance + Goal Seek” Method

In lecture for this course, balances are written in a residual form and solved by Goal Seek rather than by algebraic rearrangement. The workflow has two key ideas:

1. **Create an ideal reference outlet for turbines/pumps using the entropy balance.** For an adiabatic, reversible steady-flow device, $\dot{S}_{gen} = 0$ and $\sum \dot{Q}/T = 0$. With one inlet and one outlet:

$$0 = \dot{m}(s_{in} - s_{out,ideal}) \Rightarrow s_{out,ideal} = s_{in}.$$

In Excel, we guess a state variable (typically quality x for two-phase conditions, or temperature T for superheated/subcooled conditions) and Goal Seek the entropy residual to zero.

2. **Compute the real outlet by enforcing the efficiency equation (again via Goal Seek).** The ideal outlet defines an ideal enthalpy drop/rise. The efficiency equation then determines the real enthalpy drop/rise and therefore the real outlet state.

1.3.1 Mass Balance (steady-flow)

General steady-flow mass balance:

$$\sum \dot{m}_{in} - \sum \dot{m}_{out} = 0 \quad (1)$$

1.3.2 Energy Balance (steady-flow, negligible KE/PE)

General steady-flow energy balance:

$$\sum \dot{Q} - \sum \dot{W}_s + \sum \dot{m}_{in} h_{in} - \sum \dot{m}_{out} h_{out} = 0 \quad (2)$$

1.3.3 Boiler Energy Balance (no shaft work)

For the boiler (no shaft work on the control volume, negligible KE/PE):

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{Q}_B \quad (3)$$

1.3.4 Turbine Energy Balance (adiabatic)

For an adiabatic turbine (one inlet, one outlet, negligible KE/PE):

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{W}_t \quad (4)$$

1.3.5 Pump Energy Balance (adiabatic)

For an adiabatic pump (one inlet, one outlet, negligible KE/PE):

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{W}_p \quad (5)$$

1.3.6 Condenser Energy Balance (no shaft work)

For the condenser (no shaft work, negligible KE/PE):

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{Q}_C \quad (6)$$

1.3.7 Open Feedwater Heater (OFWH) Mass Balance

For the open feedwater heater (mixer):

$$\dot{m}_8 = \dot{m}_3 + \dot{m}_7 \quad (7)$$

1.3.8 Open Feedwater Heater (OFWH) Energy Balance (adiabatic mixer)

For the OFWH (adiabatic, no shaft work):

$$0 = \dot{m}_3 h_3 + \dot{m}_7 h_7 - \dot{m}_8 h_8 \quad (8)$$

1.3.9 Entropy Balance (general form)

General steady-flow entropy balance:

$$\sum \dot{m}_{out} s_{out} - \sum \dot{m}_{in} s_{in} - \sum_k \frac{\dot{Q}_k}{T_k} = \dot{S}_{gen} \geq 0 \quad (9)$$

1.3.10 Ideal (reversible, adiabatic) Outlet Condition

For an ideal turbine/pump step ($\dot{Q} = 0$ and $\dot{S}_{gen} = 0$):

$$s_{out,ideal} = s_{in} \quad (10)$$

1.3.11 Turbine Isentropic Efficiency

For turbines:

$$\eta_t = \frac{h_{in} - h_{out}}{h_{in} - h_{out,ideal}} \quad (11)$$

1.3.12 Pump Isentropic Efficiency

For pumps (work input device):

$$\eta_p = \frac{h_{out,ideal} - h_{in}}{h_{out} - h_{in}} \quad (12)$$

1.3.13 Thermal Efficiency

Cycle thermal efficiency:

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_B} \quad (13)$$

1.3.14 Maximum Theoretical (Carnot) Efficiency

For a heat engine operating between a hot reservoir at T_H and the environment (cold reservoir) at T_0 :

$$\eta_{max} = \eta_{Carnot} = 1 - \frac{T_0}{T_H} \quad (14)$$

1.3.15 Thermodynamic Efficiency

In this assignment, the Excel sheet defines thermodynamic efficiency as:

$$\eta_{td} = \frac{\dot{W}_{net}}{\dot{W}_{net} + \dot{W}_{lost,tot}} \quad (15)$$

1.4 Lost Work / Exergy Destruction

Lost work (exergy destruction) quantifies irreversibility. In availability form, a common steady-flow expression is:

$$\dot{W}_{lost} = \sum \dot{B}_{in} - \sum \dot{B}_{out} + \sum_k \left(1 - \frac{T_0}{T_k}\right) \dot{Q}_k - \dot{W}_s \quad (16)$$

where $\dot{B} = \dot{m}(h - T_0s)$ is the stream availability rate (availability flow), T_0 is the environment temperature, and $\dot{W}_s$ is the shaft work *leaving* the control volume. In this homework, the spreadsheet uses $T_0 = 300$ K. The lost work values reported here are those computed directly by the workbook from availability differences and the device interactions.

1.5 Assumptions and Given Data

- Steady-state, steady-flow.
- Negligible KE/PE changes.
- Turbines: $\eta_t = 0.85$ (both turbines).
- Pumps: $\eta_p = 0.75$ (both pumps).
- Environment: $T_0 = 300$ K.
- “Hot reservoir” for max theoretical efficiency: $T_H = 1250$ K.
- Operation: 8000 hr/yr.
- Boiler efficiency: $\eta_B = 0.85$.
- Methane HHV: 55.5 MJ/kg.
- Utility prices:

Electricity 0.0775 \$/kWh

Cooling Water 0.354 \$/GJ,

Natural gas 4.25 \$/GJ.

- Air composition: $y_{O_2} = 0.21$, $y_{N_2} = 0.79$; dry flue gas $y_{O_2} = 0.03$.
- NO_x factor: 0.05 lb/MMBtu (HHV).
- Worldwide fossil-fuel CO_2 emissions (NOAA Climate.gov, citing Global Carbon Budget 2024): 37.4 Gt CO_2 /yr.

1.6 Note on Quality x and Superheated States

Quality x is only defined in the two-phase region (liquid–vapor). If a Goal Seek step produces a computed “quality” greater than 1, the outlet is **superheated vapor**. In another case if the computed “quality” is less than 0, the outlet is **subcooled liquid**. In these cases, the correct workflow is:

1. set $x = 1$ (superheated vapor), or set $x = 0$ (subcooled liquid)
2. switch the adjustable state variable from x to **temperature**,
3. Goal Seek the temperature until the turbine efficiency balance is satisfied.

2 System Definition and State Table

2.1 Cycle Diagram and Labels

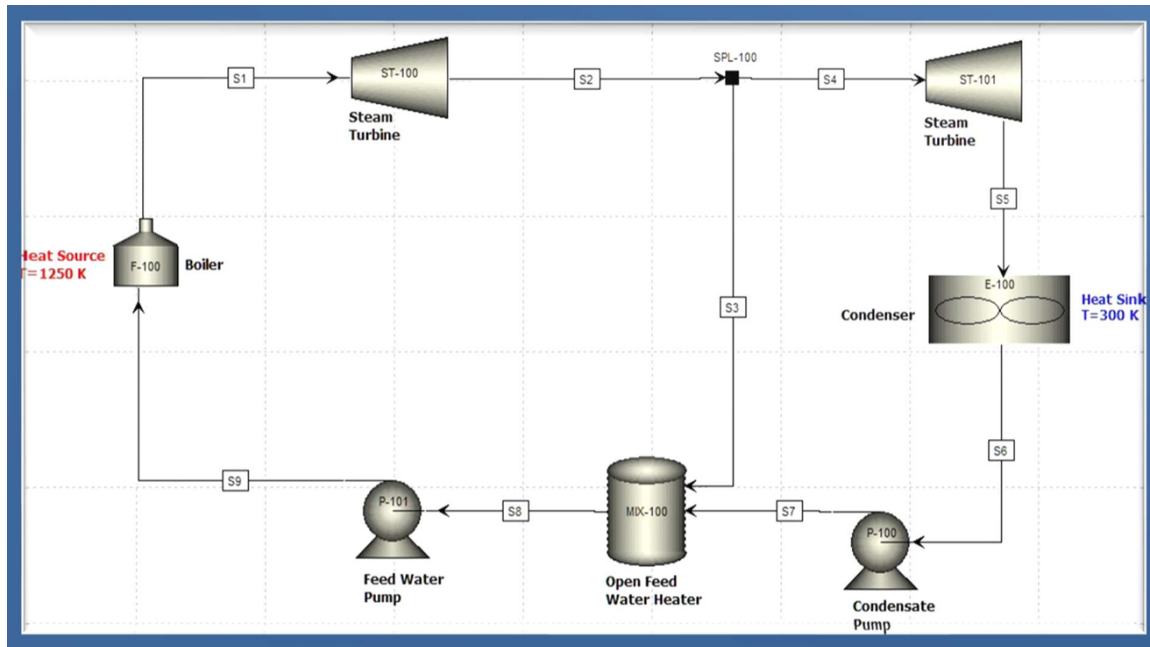


Figure 1: Regenerative Rankine Cycle

2.2 Thermodynamic State Table

Table 1: Thermodynamic States (from Excel).

State	P (bar)	T ($^{\circ}\text{C}$)	x	h (kJ/kg)	s (kJ/(kg K))
S1	40.000	450.000	1.000	3331.219	6.93882
S2	2.000	135.021	1.000	2737.419	7.20451
S3	2.000	135.021	1.000	2737.419	7.20451
S4	2.000	135.021	1.000	2737.419	7.20451
S5	0.0738	40.000	0.894	2318.071	7.44227
S6	0.0738	40.000	0.000	167.462	0.57208
S7	2.000	40.022	0.000	167.724	0.57230
S8	2.000	119.231	0.000	500.479	1.51952
S9	40.000	119.870	0.000	505.865	1.52300

3 Detailed Unit Analyses

3.1 Solve Strategy

1. **Start at S1 (boiler outlet):** $P_1 = 40 \text{ bar}$, $T_1 = 450^\circ\text{C}$, and guess steam flow rate as 1 kmol/s .
2. **HP turbine ideal outlet (S2 ideal):** set discharge pressure to 2 bar and enforce $s_{2,ideal} = s_1$ by Goal Seeking a guessed quality $x_{2,ideal}$ (two-phase assumption) until the entropy residual is zero.
3. **HP turbine real outlet (S2):** apply the turbine efficiency balance Eq. (11). If Goal Seek produces $x_2 > 1$, then S2 is superheated; use the workflow in §1.6.
4. **Split at S2:** define b as the bleed fraction, then $\dot{m}_{S3} = b\dot{m}_{S1}$ and $\dot{m}_{S4} = (1 - b)\dot{m}_{S1}$.
5. **LP turbine ideal outlet (S5 ideal):** set $s_{5,ideal} = s_4$ at condenser pressure via Goal Seek (quality if two-phase).
6. **LP turbine real outlet (S5):** enforce Eq. (11) for the LP turbine (Goal Seek on quality or temperature).
7. **Condenser (S5 $\rightarrow$ S6):** set S6 to saturated liquid at the condenser pressure (so $x_6 = 0$).
8. **Condensate pump (S6 $\rightarrow$ S7 ideal $\rightarrow$ S7):** enforce $\Delta s = 0$ for ideal, then Eq. (12) for the real pump.
9. **OFWH (mixer) (S3 + S7 $\rightarrow$ S8):** choose the outlet requirement ($T_{sat}(P) - 1^\circ\text{C}$) and Goal Seek b until the mixer energy balance is exactly satisfied.
10. **Feedwater pump (S8 $\rightarrow$ S9 ideal $\rightarrow$ S9):** enforce $\Delta s = 0$ for ideal, then Eq. (12) for the real pump.
11. **Close the loop (S9 $\rightarrow$ S1 through boiler):** compute boiler duty from the enthalpy rise.
12. **Finally, scale the main steam flow until net power is 500 kW.** (Goal Seek the steam flow cell until the overall power balance residual is zero.)

3.2 Boiler

3.2.1 Energy Balance

For the boiler (steady-flow, negligible KE/PE, no shaft work on the control volume):

$$0 = \dot{m}(h_{in} - h_{out}) + \dot{Q}_B \quad (17)$$

Thus,

$$\dot{Q}_B = \dot{m}(h_{out} - h_{in})$$

From the spreadsheet (Duty row for S1 column):

$$\dot{Q}_B = 1482.359 \text{ kW}.$$

With boiler efficiency $\eta_B = 0.85$, the fuel heat input rate is

$$\dot{Q}_{fuel} = \frac{\dot{Q}_B}{\eta_B} = \frac{1482.359}{0.85} = 1743.952 \text{ kW}.$$

3.3 High-Pressure Turbine (ST-100): S1 $\rightarrow$ S2

3.3.1 Ideal outlet construction (entropy balance)

Ideal turbine outlet at the discharge pressure satisfies

$$s_{2,ideal} = s_1.$$

In the workbook, this was enforced by Goal Seek on an assumed outlet variable (quality if two-phase, temperature if superheated) until the entropy residual cell ("Entropy Balance Turbine") is 0.

From the state table (Excel):

$$h_1 = 3331.219 \text{ kJ/kg}, \quad h_{2,ideal} = 2632.331 \text{ kJ/kg}.$$

3.3.2 Real outlet (efficiency balance)

Using Eq. (11):

$$\eta_t = 0.85 = \frac{h_1 - h_2}{h_1 - h_{2,ideal}}.$$

In Excel, the real outlet state was obtained by Goal Seek on the turbine efficiency residual until it equaled zero.

From the state table (Excel):

$$h_2 = 2737.419 \text{ kJ/kg.}$$

If the computed real quality tries to exceed 1, quality is not the correct state variable; S2 is superheated and T_2 must be Goal Sought instead (see §1.6).

3.3.3 Power

The turbine energy balance (adiabatic) is:

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{W}_t \quad (18)$$

so the turbine power output magnitude is

$$|\dot{W}_{t,HP}| = \dot{m} (h_1 - h_2).$$

Using the Excel mass flow $\dot{m} = 0.524659 \text{ kg/s}$ and the enthalpy drop:

$$|\dot{W}_{t,HP}| = 0.524659 (3331.219 - 2737.419) = 311.542 \text{ kW.}$$

Spreadsheet “Duty [Power]” gives:

$$\dot{W}_{t,HP} = -311.542 \text{ kW}$$

(negative sign in the sheet indicates *power produced* by the turbine under that sign convention).

3.4 Low-Pressure Turbine (ST-101): S4 → S5

3.4.1 Ideal outlet construction (entropy balance)

Ideal turbine outlet at the discharge pressure satisfies

$$s_{5,ideal} = s_4.$$

In the workbook, this was enforced by Goal Seek on an assumed outlet variable (quality if two-phase, temperature if superheated) until the entropy residual cell is 0.

From the state table (Excel):

$$h_4 = 2737.419 \text{ kJ/kg}, \quad h_{5,ideal} = 2244.208 \text{ kJ/kg}.$$

3.4.2 Real outlet (efficiency balance)

Using Eq. (11) with $\eta_t = 0.85$:

$$\eta_t = \frac{h_4 - h_5}{h_4 - h_{5,ideal}}$$

In Excel, the real outlet state was obtained by Goal Seek on the turbine efficiency residual until it equaled zero.

From the state table (Excel):

$$h_5 = 2318.071 \text{ kJ/kg}.$$

3.4.3 Power

Only the $(1 - b)$ fraction of the main steam flow expands through ST-101, so

$$\dot{m}_4 = (1 - b)\dot{m}_1.$$

Using $b = 0.12949$ and $\dot{m}_1 = 0.524\,659 \text{ kg/s}$:

$$\dot{m}_4 = (1 - 0.12949)0.524\,659 = 0.456\,728 \text{ kg/s}.$$

The turbine power output magnitude is

$$|\dot{W}_{t,LP}| = \dot{m}_4 (h_4 - h_5) = 0.456\,728 (2737.419 - 2318.071) = 191.403 \text{ kW}.$$

Spreadsheet “Duty [Power]” gives:

$$\dot{W}_{t,LP} = -191.403 \text{ kW}.$$

(negative sign in the sheet indicates *power produced* by the turbine under that sign convention).

3.5 Condenser: S5 → S6

Condenser has no shaft work; energy balance:

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{Q}_C \quad (19)$$

Spreadsheet duty:

$$\dot{Q}_C = -982.359 \text{ kW}$$

(heat rejected from the working fluid).

3.6 Condensate Pump (P-100): S6 → S7

3.6.1 Ideal outlet construction (entropy balance)

For the ideal (reversible, adiabatic) pump,

$$s_{7,ideal} = s_6.$$

In the workbook, this was enforced by Goal Seek until the pump entropy residual is zero.

From the state table (Excel):

$$h_6 = 167.462 \text{ kJ/kg}, \quad h_{7,ideal} = 167.658 \text{ kJ/kg}.$$

3.6.2 Real outlet (pump efficiency balance)

Using the pump isentropic efficiency Eq. (12) with $\eta_p = 0.75$:

$$\eta_p = \frac{h_{7,ideal} - h_6}{h_7 - h_6}.$$

In Excel, the real outlet state was obtained by Goal Seek on the pump efficiency residual until it equaled zero.

From the state table (Excel):

$$h_7 = 167.724 \text{ kJ/kg}.$$

3.6.3 Pump Power

The pump energy balance (adiabatic) is:

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{W}_p \quad (20)$$

so the pump power input magnitude is

$$|\dot{W}_{p,100}| = \dot{m} (h_{out} - h_{in}).$$

Spreadsheet “Duty [Power]” gives:

$$\dot{W}_{p,100} = 0.119\,388 \text{ kW}.$$

(positive sign in the sheet indicates *power required* by the pump under that sign convention).

3.7 Open Feedwater Heater (OFWH / Mixer): S3 + S7 → S8

3.7.1 Mass balance

Let b be the bleed fraction from S2:

$$\dot{m}_3 = b\dot{m}_1, \quad \dot{m}_4 = (1 - b)\dot{m}_1.$$

Mixer total mass balance:

$$\dot{m}_8 = \dot{m}_3 + \dot{m}_7.$$

3.7.2 Energy balance

$$0 = \dot{m}_3 h_3 + \dot{m}_7 h_7 - \dot{m}_8 h_8 \quad (21)$$

The outlet condition was enforced as 1 °C subcooled at the OFWH pressure, producing $T_8 = 119.231$ °C at 2 bar. Goal Seek was used on b until Eq. (21) residual is zero. Final:

$$b = 0.12949.$$

3.8 Feedwater Pump (P-101): S8 → S9

3.8.1 Ideal outlet construction (entropy balance)

For the ideal (reversible, adiabatic) pump,

$$s_{9,ideal} = s_8.$$

In the workbook, this was enforced by Goal Seek until the pump entropy residual is zero.

From the state table (Excel):

$$h_8 = 500.479 \text{ kJ/kg}, \quad h_{9,ideal} = 504.519 \text{ kJ/kg}.$$

3.8.2 Real outlet (pump efficiency balance)

Using the pump isentropic efficiency Eq. (12) with $\eta_p = 0.75$:

$$\eta_p = \frac{h_{9,ideal} - h_8}{h_9 - h_8}.$$

In Excel, the real outlet state was obtained by Goal Seek on the pump efficiency residual until it equaled zero.

From the state table (Excel):

$$h_9 = 505.865 \text{ kJ/kg}.$$

3.8.3 Pump Power

The pump energy balance (adiabatic) is:

$$0 = \dot{m} (h_{in} - h_{out}) + \dot{W}_p \quad (22)$$

so the pump power input magnitude is

$$|\dot{W}_{p,101}| = \dot{m} (h_{out} - h_{in}).$$

Spreadsheet “Duty [Power]” gives:

$$\dot{W}_{p,101} = 2.825 \text{ 627 kW}.$$

(positive sign in the sheet indicates *power required* by the pump under that sign convention).

4 Overall Cycle Balances and Key Performance Metrics

4.1 Overall Power Balance (The “Reality Check”)

Net power is defined (physical sign convention) as

$$\dot{W}_{net} = |\dot{W}_{t,HP}| + |\dot{W}_{t,LP}| - (\dot{W}_{p,100} + \dot{W}_{p,101}) .$$

In the sheet, the power-balance residual is driven to zero by Goal Seek on the main steam flow. Final solution satisfies:

$$\dot{W}_{net} = 500 \text{ kW}.$$

4.2 Steam Flow Rate for 500 kW

From the ENG sheet:

$$\dot{V}_{steam} = 2.105 \text{ MMSCFD}$$

(Internal sheet values: $\dot{n}_{S1} = 0.029123 \text{ kmol/s}$, $\dot{m}_{S1} = 0.524659 \text{ kg/s}$.)

4.3 Thermal Efficiency

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_B}.$$

Using $\dot{W}_{net} = 500 \text{ kW}$ and $\dot{Q}_B = 1482.359 \text{ kW}$:

$$\eta_{th} = 0.3373 = 33.73\%$$

4.4 Maximum Theoretical (Carnot) Efficiency (as assigned)

Using the assigned temperatures:

$$\eta_{\max} = 1 - \frac{T_0}{T_H} = 1 - \frac{300}{1250} = 0.760.$$

$$\eta_{\max} = 76.0\%$$

4.5 Thermodynamic Efficiency

In this course the thermodynamic (second-law) efficiency for a power-producing cycle is defined as the fraction of the available work that is actually delivered as net power:

$$\eta_{td} = \frac{\dot{W}_{\text{net}}}{\dot{W}_{\text{net}} + \dot{W}_{\text{lost},tot}}.$$

Using $\dot{W}_{\text{net}} = 500 \text{ kW}$ and $\dot{W}_{\text{lost},tot} = 626.593 \text{ kW}$:

$$\eta_{td} = \frac{500}{500 + 626.593} = 0.4438 = 44.38\%$$

5 Lost Work (Exergy Destruction) Results

Table 2: Lost work (exergy destruction) by unit (kW).

Unit	$\dot{W}_{lost}$ (kW)	Interpretation
Boiler	496.711	Largest irreversibility due to heat addition at finite ΔT and combustion/boiler losses.
HP Turbine (ST-100)	41.820	Internal irreversibility represented by $\eta_t < 1$.
LP Turbine (ST-101)	32.359	Additional turbine irreversibility during low-pressure expansion.
Condenser	41.252	Heat rejection near the environment creates entropy generation.
Condensate Pump (P-100)	0.029	Small because liquid pumping work is small.
Feedwater Pump (P-101)	0.545	Larger than P-100 due to larger pressure rise to boiler pressure.
OFWH / Water Heater (Mixer)	13.877	Mixing irreversibility (two streams at same pressure but different states).
Total	626.593	Total exergy destruction in the cycle devices.

5.1 Discussion

The boiler is the dominant contributor to lost work because heat addition (and combustion/boiler inefficiency) creates significant entropy generation. The turbines are next because mechanical/flow irreversibilities are captured through $\eta_t < 1$. The condenser contributes because rejecting heat to a near-environment sink carries low exergy but still produces entropy. Pumps are small contributors. The OFWH has a nonzero loss because direct mixing increases entropy even if the device is adiabatic.

6 Utilities and Costing

6.1 Cooling Water Requirement (calculated from condenser duty)

Using $|\dot{Q}_C| = 982.359 \text{ kW}$, $c_p \approx 4.18 \text{ kJ}/(\text{kg K})$, and $\Delta T = 10 \text{ K}$:

$$\dot{m}_{cw} = \frac{|\dot{Q}_C|}{c_p \Delta T} = \frac{982.359}{4.18 \times 10} = 23.50 \text{ kg/s}.$$

6.2 Annual Utility Costs

Pump electricity: Total pump power:

$$\dot{W}_{p,tot} = \dot{W}_{p,100} + \dot{W}_{p,101} = 0.119388 + 2.825627 = 2.945015 \text{ kW}.$$

Annual electricity use:

$$E_{pump} = \dot{W}_{p,tot} \times (8000 \text{ hr/yr}) = 23560.12 \text{ kWh/yr}.$$

At 0.0775 \$/kWh:

$$C_{pump} = 1825.91 \text{ \$/yr}.$$

Cooling water: Spreadsheet reports cooling duty:

$$Q_{cool,yr} = 28291.94 \text{ GJ/yr}.$$

At 0.354 \$/GJ:

$$C_{cw} = 10015.35 \text{ \$/yr}.$$

Natural gas: Fuel energy input rate is $\dot{Q}_{fuel} = 1743.952 \text{ kW}$. Spreadsheet reports annual fuel energy:

$$Q_{fuel,yr} = 50226.08 \text{ GJ/yr}.$$

At 4.25 \$/GJ:

$$C_{NG} = 213459.68 \text{ \$/yr}.$$

Table 3: Annual cost breakdown (ENG sheet values).

Component	Annual Basis	Cost (\$/yr)
Pump electricity	23 560 kWh/yr	1825.91
Cooling water	28 291.94 GJ/yr	10015.35
Natural gas	50 226.08 GJ/yr	213459.68
Total		225300.94

$$C_{annual} = 2.253 \times 10^5 \text{ \$/yr}$$

7 Optimization Study (HP Turbine Discharge Pressure)

7.1 Objective (what is being minimized)

The decision variable is the HP turbine discharge pressure, P_2 (which also sets the OFWH pressure in this setup). The objective is to minimize the total annual operating cost:

$$C_{annual}(P_2) = C_{NG}(P_2) + C_{cw}(P_2) + C_{pump}(P_2).$$

Because P_2 affects the turbine outlet states, bleed condition, boiler duty, condenser duty, and pump duties, it ultimately changes the annual costs.

7.2 Method Used (Pressure sweep + polynomial fit + derivative test)

Rather than using a process simulator, I performed the optimization directly in Excel using a **pressure sweep** and a **curve-fit/derivative** approach:

1. I swept P_2 from 1 to 10 bar (integer steps).
2. For each P_2 value, I re-solved the cycle using the same Goal Seek sequence used for the base case:
 - ST-100 ideal: entropy residual $\rightarrow 0$,
 - ST-100 real: turbine efficiency residual $\rightarrow 0$,
 - ST-101 ideal: entropy residual $\rightarrow 0$,
 - ST-101 real: turbine efficiency residual $\rightarrow 0$,
 - pumps ideal/real similarly,
 - OFWH: mixer energy residual $\rightarrow 0$ (vary b),
 - overall net power residual $\rightarrow 0$ (vary main steam flow to keep $\dot{W}_{net} = 500$ kW).
3. After each solve, I recorded the resulting total annual cost C_{annual} .
4. I fit the (P_2, C_{annual}) data to a 6th-order polynomial (Excel trendline).
5. I differentiated the polynomial to obtain $C'(P_2)$ and used Excel to find the critical point where $C'(P_2) = 0$.
6. I evaluated $C''(P_2)$ at that critical point. Since $C''(P_{2,opt}) > 0$, the curve is concave up and the point is a minimum.

7.3 Sweep Data (1–10 bar)

Table 4: Recorded annual cost versus HP discharge pressure P_2 (Excel sweep).

P_2 (bar)	C_{annual} (\$/yr)
1.0000	226330.2675
2.0000	225233.0045
3.0000	224333.7099
4.0000	223801.6535
5.0000	223714.7461
6.0000	223940.5146
7.0000	224204.8320
8.0000	224504.8959
9.0000	225048.6168
10.0000	225396.4850

7.4 Best-fit equation, derivative, and second derivative

Using an Excel polynomial trendline (degree 6), the best-fit annual cost model was:

$$C(P_2) = -0.1899P_2^6 + 6.6439P_2^5 - 91.248P_2^4 + 608.64P_2^3 - 1855.9P_2^2 + 1411.1P_2 + 226247. \quad (23)$$

Taking the derivative (done algebraically, then implemented in Excel):

$$\begin{aligned} C'(P_2) &= 6(-0.1899)P_2^5 + 5(6.6439)P_2^4 + 4(-91.248)P_2^3 + 3(608.64)P_2^2 + 2(-1855.9)P_2 + 1411.1 \\ &= -1.1394P_2^5 + 33.2195P_2^4 - 364.992P_2^3 + 1825.92P_2^2 - 3711.8P_2 + 1411.1. \end{aligned} \quad (24)$$

Taking the second derivative:

$$\begin{aligned} C''(P_2) &= 5(-1.1394)P_2^4 + 4(33.2195)P_2^3 + 3(-364.992)P_2^2 + 2(1825.92)P_2 - 3711.8 \\ &= -5.6970P_2^4 + 132.878P_2^3 - 1094.976P_2^2 + 3651.84P_2 - 3711.8. \end{aligned} \quad (25)$$

7.5 How Excel found the critical point (root of the first derivative)

To locate the optimum, Excel was used to find the **root of the first derivative**:

$$C'(P_2) = 0.$$

In the spreadsheet, P_2 was placed in a single cell (e.g., x), the derivative formula $C'(x)$ was computed in another cell, and **Goal Seek** was applied:

- **Set cell:** the cell containing $C'(x)$
- **To value:** 0
- **By changing cell:** the cell containing $x = P_2$

This produced the critical point:

$$P_{2,opt} = 4.6909 \text{ bar}$$

7.6 Second-derivative test (confirming a minimum)

After finding $P_{2,opt}$, the value was substituted into the second derivative $C''(P_2)$ (computed in Excel as well):

$$C''(P_{2,opt}) = 281.4938 > 0.$$

Since $C''(P_{2,opt})$ is positive, the curve is **concave up** ("smile-shaped") at the critical point, so the critical point is a **local minimum**.

concave up (minimum): $\begin{matrix} + & + \\ & \smile \end{matrix}$

7.7 Plot

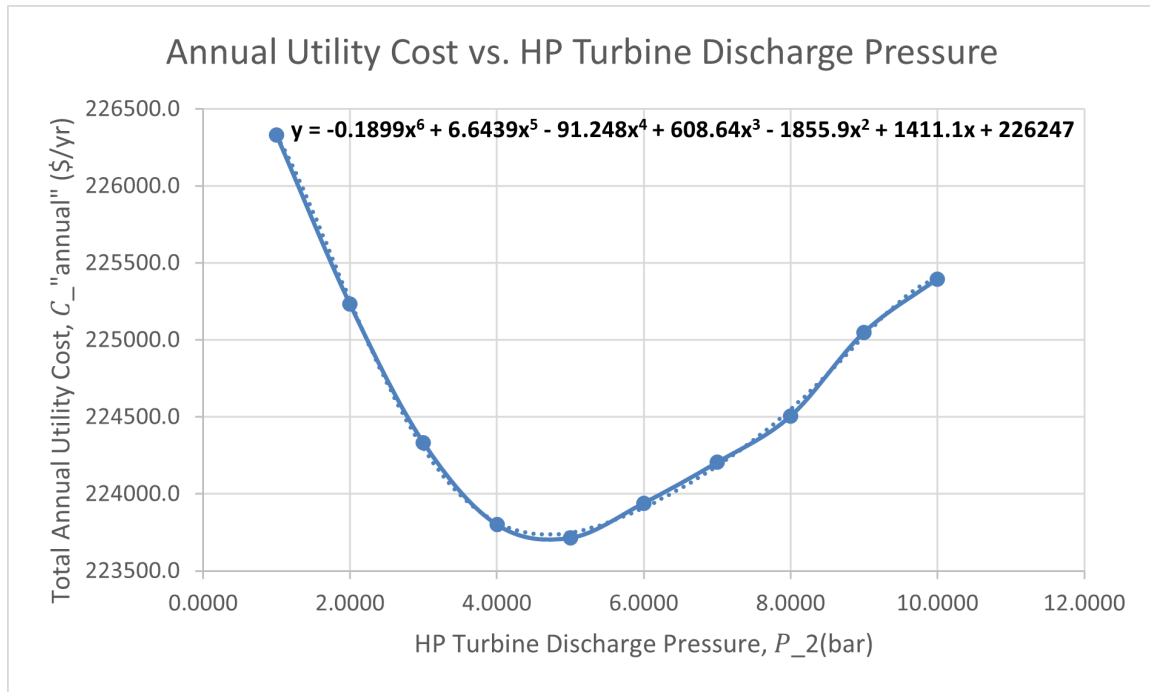


Figure 2: Annual cost C_{annual} versus HP turbine discharge pressure P_2 . Points are Excel sweep data (Table 4); curve is the polynomial fit Eq. (23). The minimum occurs at $P_{2,opt} \approx 4.69$ bar.

7.8 Final reported optimum

At the optimum, the fitted model and/or Excel evaluation gives:

$$C_{annual}(P_{2,opt}) = \$223737.5249/\text{yr}.$$

$$P_{2,opt} = 4.6909 \text{ bar}, \quad C_{annual,min} \approx \$2.237 \times 10^5/\text{yr}$$

8 Combustion, Air Requirement, and Emissions

8.1 Fuel rate from boiler heat requirement

Fuel energy input:

$$\dot{Q}_{fuel} = 1743.952 \text{ kW} = 1743.952 \text{ kJ/s.}$$

With HHV = 55.5 MJ/kg = 55 500 kJ/kg, methane mass flow:

$$\dot{m}_{CH_4} = \frac{\dot{Q}_{fuel}}{HHV} = \frac{1743.952}{55500} = 0.03143 \text{ kg/s.}$$

8.2 CO₂ emissions

For complete combustion: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, so 1 mol CO₂ per 1 mol CH₄.

$$\dot{m}_{CO_2} = 0.0862 \text{ kg/s,} \quad m_{CO_2, yr} = 2.0 \times 10^{-6} \text{ Gt/yr.}$$

$$m_{CO_2} = 2.0 \times 10^{-6} \text{ GtCO}_2/\text{yr}$$

8.3 Air requirement (3% O₂ dry flue gas)

The ENG sheet reports air flow:

$$\dot{V}_{air} = 1.5495 \text{ MMSCFD}$$

8.4 NO_x emissions

With NO_x factor 0.05 lb/MMBtu and annual fuel energy, the ENG sheet reports:

$$m_{NO_x} = 1079.66 \text{ kg/yr.}$$

$$m_{NO_x} = 1.08 \times 10^3 \text{ kg/yr}$$

8.5 Percent of worldwide CO₂

Using worldwide fossil-fuel CO₂ emissions (reported as 37.4 GtCO₂/yr for 2024):

$$\%world = \frac{2.0 \times 10^{-6}}{37.4} \times 100 \approx 5.3 \times 10^{-6}\%.$$

9 Summary of Required Results

Table 5: Summary of key results (from ENG sheet).

Quantity	Value
Steam Flow Rate (MMSCFD)	2.105
Bleed Fraction (b)	0.12949
Thermal Efficiency (%)	33.73
Maximum Theoretical Efficiency (%)	76.0
Thermodynamic Efficiency (%)	44.38
Total Lost Work (kW)	626.593
Annual Utilities Cost (\$/yr)	225300.94
Optimum HP Discharge Pressure (bar)	4.69
Air Flow (MMSCFD)	1.5495
CO ₂ Emissions (GtCO ₂ /yr)	2.0×10^{-6}
NO _x Emissions (kg/yr)	1.08×10^3

10 Conclusion

- A regenerative Rankine cycle with an OFWH was solved using the instructor “ideal outlet + efficiency balance + untouched residuals” method.
- The net power constraint of 500 kW was enforced by scaling steam flow, yielding $\dot{V}_{steam} = 2.105$ MMSCFD.
- The bleed fraction required to obtain a 1 °C subcooled OFWH outlet at 2 bar was $b = 0.12949$.
- The cycle thermal efficiency is 33.73%, while the assigned theoretical upper bound is 76.0%, giving a thermodynamic efficiency of 44.38%.
- Total exergy destruction is 626.59 kW, dominated by the boiler, followed by turbines and condenser.
- The annual operating cost is approximately $\$2.25 \times 10^5$ /yr, dominated by natural gas.
- Optimization of the HP turbine discharge pressure can be completed via a P_2 sweep with repeated Goal Seek solves. However, running many sweep points requires repeated Goal Seek solves and can be time-consuming.

References

1. CHE 301 course Excel property functions (HSteam/SSeam) and your ENG Calculations worksheet (HW#02 workbook).
2. Lindsey, R. (2025, May 21). *Climate Change: Atmospheric Carbon Dioxide* (J. Miller, Ed.). NOAA Climate.gov; National Oceanic and Atmospheric Administration.
<https://www.climate.gov/news-features/understanding-climate/climate-change-atmospheric-carbon-dioxide>

Appendices

A How to Reproduce the Goal Seek Steps (Checklist).

Use this exact order and reference appendix B to follow my workflow:

1. **ST-100 ideal:** Goal Seek the *entropy residual* $\rightarrow 0$ by changing the **outlet quality x** (light blue, Cell #1).
2. **ST-100 real:** Goal Seek the *turbine efficiency residual* $\rightarrow 0$ by changing the **outlet quality x** (light blue, Cell #2).
3. **ST-101 ideal:** Goal Seek the *entropy residual* $\rightarrow 0$ by changing the **outlet quality x** (blue, Cell #1).
4. **ST-101 real:** Goal Seek the *turbine efficiency residual* $\rightarrow 0$ by changing the **outlet quality x** (blue, Cell #2).
5. **P-100 ideal:** Goal Seek the *entropy residual* $\rightarrow 0$ by changing the **outlet temperature T** (light green, Cell #1).
6. **P-100 real:** Goal Seek the *pump efficiency residual* $\rightarrow 0$ by changing the **outlet temperature T** (light green, Cell #2).
7. **MIX-100 (OFWH):** Goal Seek the *mixer energy residual* $\rightarrow 0$ by changing the **bleed fraction b** .
8. **P-101 ideal:** Goal Seek the *entropy residual* $\rightarrow 0$ by changing the **outlet temperature T** (green, Cell #1).
9. **P-101 real:** Goal Seek the *pump efficiency residual* $\rightarrow 0$ by changing the **outlet temperature T** (green, Cell #2).
10. **Power balance:** Goal Seek the *overall power residual* $\rightarrow 0$ by changing the **main steam flow rate** (steam flow input cell).

B Excel Sheet (Screenshot)

Steam Properties		S1	S2 IDEAL	S2	S3	S4	S5 IDEAL	S5	S6	S7 IDEAL	S7	S8	S9 IDEAL	S9	S1
Molar Flow, kmol/s		0.0291	0.0291	0.0291	0.0038	0.0253	0.0253	0.0253	0.0253	0.0253	0.0253	0.0291	0.0291	0.0291	0.0291
Mass Flow, kg/s		0.5245	0.5245	0.5245	0.0679	0.4566	0.4566	0.4566	0.4566	0.4566	0.4566	0.5245	0.5245	0.5245	0.5245
Volume Flow, m ³ /s		450.0000													450.0000
Temperature, °C		40.0000	2.0000	134.8925	134.8925	2.0000	0.0738	0.0738	0.0000	2.0000	40.0000	119.2308	119.5457	119.8619	450.0000
Pressure, bar		1.0000	0.0664	1.0000	1.0000	1.0000	0.8628	0.8935	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000
Vapor Fraction															
Molecular Weight, kg/kmol		18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150	18.0150
Molar Enthalpy, kJ/kmol		60011.902	47421.451	49309.760	49309.760	49309.760	40428.886	41761.017	3016.658	3020.190	3021.372	9015.728	9088.046	9112.169	60011.902
Molar Entropy, kJ/(kmol K)		60011.902	48753.821	49309.760	49309.760	49309.760	46377.193	46377.193	48753.821	48753.821	48753.821	50448.097	50448.097	50448.097	60011.902
Molar Density, kmol/m ³															
Enthalpy Flow, kJ/s		1747.30	1380.72	1435.70	185.92	1249.77	1024.68	1058.45	76.46	76.55	76.58	262.50	264.61	265.31	1747.30
Entropy Flow, kJ/(s K)		3.64	3.64	3.78	0.49	3.29	3.29	3.40	0.26	0.26	0.26	0.80	0.80	0.80	3.64
Availability, kJ/s		655.43	288.85	302.12	39.12	263.00	37.91	39.33	-1.91	-1.82	-1.82	23.43	25.54	25.71	655.43
Duty [Power], kJ/s [kW]			-366.58	-311.60			-225.09	-191.33	-881.99	0.09	0.12		2.11	2.81	1481.99
Efficiency				0.85				0.85			0.75			0.75	
Compressibility Factor															

Figure 3: Excel sheet used for HW#02 calculations.

C Unit Conversions Used

- $1 \text{ kW} = 1 \text{ kJ/s}$.
- $1 \text{ GJ} = 10^6 \text{ kJ}$.
- $1 \text{ MMBtu} = 1.055056 \times 10^6 \text{ kJ}$.
- $1 \text{ lb} = 0.453592 \text{ kg}$.